Information Systems in Global Business Today

Chapter 1 & 2 — Full Depth Study Notes

1. What is an Information System?

From the notes: Information + Data + Activities □ Management

This is not just a formula. It represents the core working of any organization.

> Definition: A system that collects data, processes it, and produces meaningful information to support management and decision-making.

Data

Raw facts with no meaning on their own:

- Numbers
- Records
- Transactions

Examples: 500 units sold. 10 employees absent.

Information

Processed data with meaning and purpose:

- "Sales increased by 20%"
- "Attendance dropped this week"

The 3 Core Activities

Your notes highlight three activities that transform data into information:

1. Input

Collecting raw data from the environment.

Example: Customer order entered into the system.

2. Processing

Converting raw data into a meaningful form.

- Calculating total bill
- Updating inventory records

3. Output

Delivering useful information to users.

- Invoice generated
- Report produced

> Final Connection: These 3 activities exist to support management decisions.
> Data □ Input □ Processing □ Output □ Information □ Management.

2. Levels in an Organization

Every organization has three levels, each with a different focus and information need.

1. Operational Level

The lowest level. Focused on daily, routine work.

- Recording sales
- Processing payroll
- Handling daily transactions

Systems used: TPS (Transaction Processing System)

2. Middle Management

Focused on monitoring, controlling, and coordinating.

- Monthly performance reports
- Budget tracking
- Department oversight

Systems used: MIS (Management Information System)

3. Senior Management

Top-level. Focused entirely on long-term strategy.

- Market expansion decisions
- New product launches
- Investment planning

Systems used: ESS (Executive Support System)

Level	Focus	Decision Type	System
Operational	Daily operations	Structured	TPS

Middle Mgmt	Monitor & control	Semi-structured	MIS / DSS
Senior Mgmt	Strategic planning	Unstructured	ESS

> Key Understanding: Different levels need different information, different systems, and different decision types.

3. Business Functions and Departments

Every organization is divided into functional departments. Each has its own role, but all are connected through information systems.

1. Sales & Marketing

- Selling products and services
- Handling customer relationships
- Running promotions

2. Finance & Accounting

- Managing money and payments
- Producing financial reports
- Budgeting and forecasting

3. Human Resources

- Hiring and onboarding employees
- Training and development
- Processing payroll

4. Production / Operations

- Manufacturing products
- Managing production schedules
- Quality control

> Important Insight: Each department has its own work but depends on the others. Information systems connect all departments into one unified operation.

4. Business Processes

> Definition: A set of connected activities that together produce a result.

A business process is not one task. It is a flow of tasks across multiple departments.

Example: Order Fulfillment

1. Sales receives the order
2. Finance checks payment and credit
3. Production prepares the product
4. Shipping delivers to customer

Process Hierarchy

From your notes: Organization □ Functions □ Processes □ Activities

- Organization — The whole company
- Functions — Departments (Sales, Finance, HR...)
- Processes — Flows of work inside and across departments
- Activities — Individual tasks within each process

> Important Insight: Information systems integrate processes across departments, making the whole organization work as one.

5. Types of Information Systems

1. Transaction Processing System (TPS)

Handles daily, routine, high-volume operations automatically.

- Sales transactions
- Payroll processing
- Booking and reservation systems

> Key Insight: TPS is the foundation of all systems.
> Without TPS □ No data. No data □ No reports. No reports □ No decisions.

2. Management Information System (MIS)

Takes data from TPS and produces structured reports for managers.

- Weekly sales report
- Employee performance summary

- Budget vs. actual spending

MIS answers: "What is happening in the business?"

3. Decision Support System (DSS)

Supports managers in making complex, semi-structured decisions through analysis and modeling.

- What happens to profit if sales double?
- What is the ROI impact if a project is delayed?
- Which pricing strategy maximizes revenue?

DSS answers: "What should we do?"

4. Executive Support System (ESS)

Designed for top-level executives. Combines internal data with external market information.

- Market trend dashboards
- Industry benchmarks
- Competitor analysis

ESS answers: "Where is the business going?"

System	Users	Purpose	Decision Type
TPS	Operational staff	Record daily transactions	Structured
MIS	Middle managers	Generate summary reports	Structured
DSS	Analysts, managers	Analyze and model scenarios	Semi-structured
ESS	Top executives	Strategic insight & trends	Unstructured

6. Enterprise Applications

Enterprise applications are large-scale systems that connect the entire organization.

1. ERP — Enterprise Resource Planning

A single unified system that connects all departments.

- Integrates Finance, HR, Production, and Sales
- Eliminates data silos — all data in one place
- Automates cross-department processes

Example: SAP, Oracle ERP

2. CRM — Customer Relationship Management

Manages all interactions with customers across the business.

- Tracks customer history and preferences
- Supports Sales, Marketing, and Customer Service
- Improves customer retention and satisfaction

Example: Salesforce, HubSpot

3. SCM — Supply Chain Management

Manages the flow of goods, information, and money across the supply chain.

- Coordinates suppliers, inventory, and delivery
- Reduces costs through optimized logistics
- Improves visibility from raw material to final delivery

> Important Insight: ERP, CRM, and SCM together integrate the entire business — from suppliers to customers.

7. Order Fulfillment Process

This is a classic example of how a single business process spans multiple departments.

1. Customer places order
2. Sales records the order in the system
3. Accounting checks credit, approves order, generates invoice
4. Production prepares the product
5. Shipping delivers the product to the customer

> Important Understanding: One process. Five departments. All connected.
> Without information systems, this flow breaks down at every handoff.

8. Payroll

Payroll is the process of calculating and distributing employee compensation.

- Calculates salary based on hours worked or fixed rate
- Applies deductions (tax, insurance, absences)

- Tracks attendance and leave records

> Important: Payroll sits at the intersection of HR and Finance.
> It is a classic TPS use case — routine, repetitive, and fully automatable.

9. Final Understanding of Chapter 1 & 2

If you truly understand this chapter, you can explain the following without memorization:

1. How a company works internally — levels, functions, processes
2. How data flows through the system — input, processing, output
3. How systems support operations — TPS, MIS, DSS, ESS, ERP, CRM, SCM
4. How departments are connected — through shared processes and shared data

> Brutally Honest Check: If you still feel you are memorizing rather than understanding —
> go back to the Order Fulfillment example and trace every step through every department.
> Understanding that one example means understanding the entire chapter.